

Srividya Manikandan

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Education

Amrita Vishwa Vidyapeetham

B.Tech in Computer Science and Engineering, GPA: 9.42

Coimbatore

07/2027 (Expected)

Amrita Vidyalayam

HSC (CBSE), Percentage: 94.8%

Coimbatore

03/2023

Amrita Vidyalayam

SSC (CBSE), Percentage: 96.2%

Coimbatore

04/2021

Experience

Metatroncube Solutions

Front-End Web Development Intern

Ontario, Canada (Online)

Internship

- Developed responsive, accessible UI components for live web projects while collaborating with developers to ensure efficient workflows and cross-browser consistency.

Amrita Vishwa Vidyapeetham

Live-in-Labs

Coimbatore

Project Work

- Conducted field research and worked with interdisciplinary teams to design conceptual and sustainable solutions addressing real-world rural challenges.

Projects

Gyroscope-based Movement Controller (STM32F4)

Embedded Systems (GitHub)

- Implemented a motion-controlled system using STM32F4 and MPU6050 gyroscope.
- Programmed GPIO and I2C at register-level with interrupt-driven data handling.

NFT-based Supply Chain Management System

Blockchain (GitHub)

- Developed Solidity smart contracts using NFT tokens for product traceability.
- Ensured secure and transparent asset transfers with ownership validation.

Fall Detection using Pose-based Features and Deep Learning

Computer Vision (GitHub)

- Extracted pose features with MediaPipe and evaluated ML models.
- Compared deep architectures to improve accuracy and robustness.

Constraint-Handled Genetic Algorithm for 0-1 Knapsack Optimization

Evolutionary Computation (GitHub)

- Enhanced GA with a repair mechanism to maintain feasible solutions.
- Improved convergence stability through parameter tuning.

Technical Skills

Areas of Interest: Computer Vision, Machine Learning, Blockchain, Embedded Systems, Theory of Computation, Genetic Algorithm

Languages & Frameworks: C, C++, Python, Java, Haskell, Solidity, JavaScript (JS), SQL, HTML, CSS, Bootstrap, React.js, Next.js, Flutter (Dart)

Publications

- **VerkChain: Fortifying Cloud Data Security with Verkle Tree-Based Blockchain Technology**

Uma J (Prof), Srividya Manikandan, Nethra Kanagaraj

- **Advancing Sustainable Seaweed Production: Linking Technology, Policy, and Ecology for a Resilient Blue Economy**

Pranaveeka R M, Manasa R, Puluputturi Nithin Reddy, Dasari Vishala, Sriharish R, Namitha Madhu, Srividya M, Niranjana K, Saravanan R (Prof), Sabarinath S (Prof)

- **Burdened by the Flow: A Gendered Perspective on Water and Sanitation in Thangachimadam, Rameshwaram, India**

Namitha Madhu, Srividya M., Niranjana K., Dr. Saravanan R (Prof), Dr. Akshay Sathish (Prof)